CIS 6261
Final Project Presentation
Option 1

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Introduction

Part 1: Defend a trained model against membership inference and adversarial examples on CIFAR-10

Part 2: Defend a model, starting from the training process, against the same attacks on CIFAR-100

Background

Membership Inference Attack: "given a machine learning model and a record, determine whether this record was used as part of the model’s training dataset or not" [6]

- Shadow models [6][5], losses [8], confidence thresholds [5], or only labels [2]

Adversarial Example Attack: “Malicious input designed to fool a machine learning model”

- Small perturbations are added to the input to change the prediction, using either model gradients [3][4] or optimization-based norm constraints [1].

Defense (Part 1)

- σ (input Gaussian noise)
- Θ (max rotation, deg)
- S (max horizontal shift, px)
- T (temperature)
- δ (temperature jitter)
- γ (tiny logit noise)
- L (EOT runs)

$$z_\ell(x) = \frac{1}{T_\ell} f\Big(g_{\theta_\ell,\Delta_\ell}(x + \eta_\ell)\Big) + \xi_\ell, \quad T_\ell = T\,(1 + \delta\, u_\ell),$$

$$\bar{z}(x) = \frac{1}{L}\sum_{\ell=1}^{L} z_\ell(x), \qquad \hat{y} = \arg\max_c \operatorname{softmax}(\bar{z}(x))_c.$$

Attacks

Attack Name	Requirements/Knowledge	Hyperparameters	Accuracy	Advantage
simple_conf_threshold_mia	logits	thresh=0.999	65.23%	0.305
simple_logits_threshold_mia	logits	thresh=9	65.43%	0.309
loss_thresh_mia	logits, target labels	thresh=0.001	65.62%	0.312
loss_proba_mia	logits, target labels, training loss distribution	thresh=0.402	65.72%	0.314
gap_attack_mia	target labels	None	57.23%	0.145
augmentation_mia	multiple queries	shifts=7, angle=9.9	58.69%	0.174
shokri_mia	target model architecture, auxillery data	num_shadow=10, num_epochs=15 [1]	58.79%	0.176

Table 1: Performance of membership inference attacks on undefended model

(b) PGD

Results - Part 1 (MIA)

σ=0.016, Θ=±5°, S=2 px, T=1.38, δ=0.05, γ=2e−4, runs=4 (val ≈ 0.82)

Attack Name	Undef. Acc.	Undef. Adv.	Defended Acc.	Defended Adv.
simple_conf_threshold_mia	65.23%	0.305	**50.59%**	**0.012**
simple_logits_threshold_mia	65.43%	0.309	**50.10%**	**0.002**
loss_thresh_mia	65.62%	0.312	**49.80%**	**-0.004**
loss_prob_mia	65.72%	0.314	**61.91%**	**0.238**
gap_attack_mia	57.23%	0.145	**57.81%**	**0.156**
augmentation_mia	58.69%	0.174	**49.71%**	**-0.006**
shokri_mia	58.79%	0.176	**60.74%**	**0.215**

Table 3: Membership inference attack performance (accuracy & advantage) before vs. after defense (runs= 4). Lower is better on the right; advantage near 0 indicates near-random performance.

Results - Part 1 (Adversarial Example)

σ=0.016, Θ=±5°, S=2 px, T=1.38, δ=0.05, γ=2e−4, runs=4 (val ≈ 0.82)

Attack	Undef. Success (%)	New Benign Acc (%)	New Adv Acc (%)	New Success (%)
Attack0 (given)	94.00	86.50	49.50	**50.50**
FGSM	77.05	83.79	30.08	**69.92**
PGD	99.71	84.47	1.56	**98.44**
CW-L2	11.13	83.30	83.20	**16.80**

Table 4: Adversarial example evaluation with the defended predictor (runs= 4). Success is defined as 100% − (adv. accuracy).

Part 2

- Trained ResNet on CIFAR-100 using HiPerGator.
- Adversarial training using examples generated with PGD.

```
def pgd_attack(model, images, labels, device, eps=8/255, alpha=2/255, iters=7):
    # Scale eps and alpha to normalized space
    eps_normalized = eps / std
    alpha_normalized = alpha / std

    for i in range(iters):
        adv_images.requires_grad = True
        loss = loss_fn(model(adv_images), labels)
        loss.backward()

        # Update with normalized alpha
        adv_images = adv_images + alpha_normalized * adv_images.grad.sign()
        # Project onto L-inf ball
        eta = torch.clamp(adv_images - original_images,
                          min=-eps_normalized, max=eps_normalized)
        adv_images = torch.clamp(original_images + eta,
                                 min=lower_bound, max=upper_bound).detach()
    return adv_images
```

```
for inputs, targets in trainloader:
    # Generate adversarial examples
    adv_inputs = pgd_attack(model, inputs, targets, device)

    # Train on adversarial examples
    optimizer.zero_grad()
    outputs = model(adv_inputs)
    loss = criterion(outputs, targets)
    loss.backward()
    optimizer.step()
```

Results - Part 2

Trained for 20 epochs (~45 minutes)

~30% Validation Accuracy

Great improvement against adversarial examples

Almost perfect against MIA

Epoch	Train Loss	Train Acc (%)	Val Acc (%)
1	4.549	2.84	7.70
5	3.966	9.60	18.29
10	3.713	13.72	23.88
15	3.547	16.38	30.12
20	3.474	17.61	30.78

Table 5: Part 2 training progress. Train accuracy is measured on adversarial examples; validation accuracy is on clean examples.

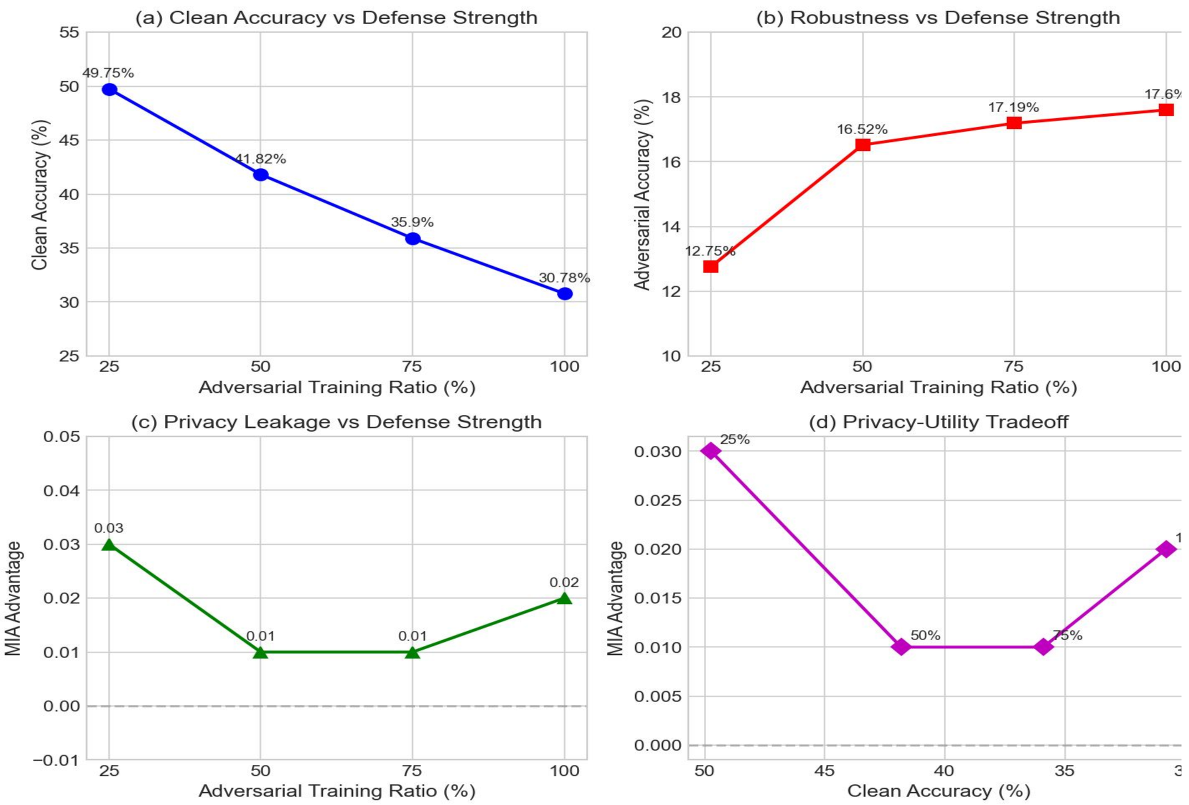
Model	Clean Acc	PGD (ϵ=8/255)	% Retained
Part 1 (standard, CIFAR-10)	87.70%	0.29%	0.3%
Part 2 (adv. trained, CIFAR-100)	30.78%	17.60%	57.2%

Table 6: Robustness comparison. Adversarial training retains 57% of clean accuracy under PGD attack, compared to near-zero for standard training.

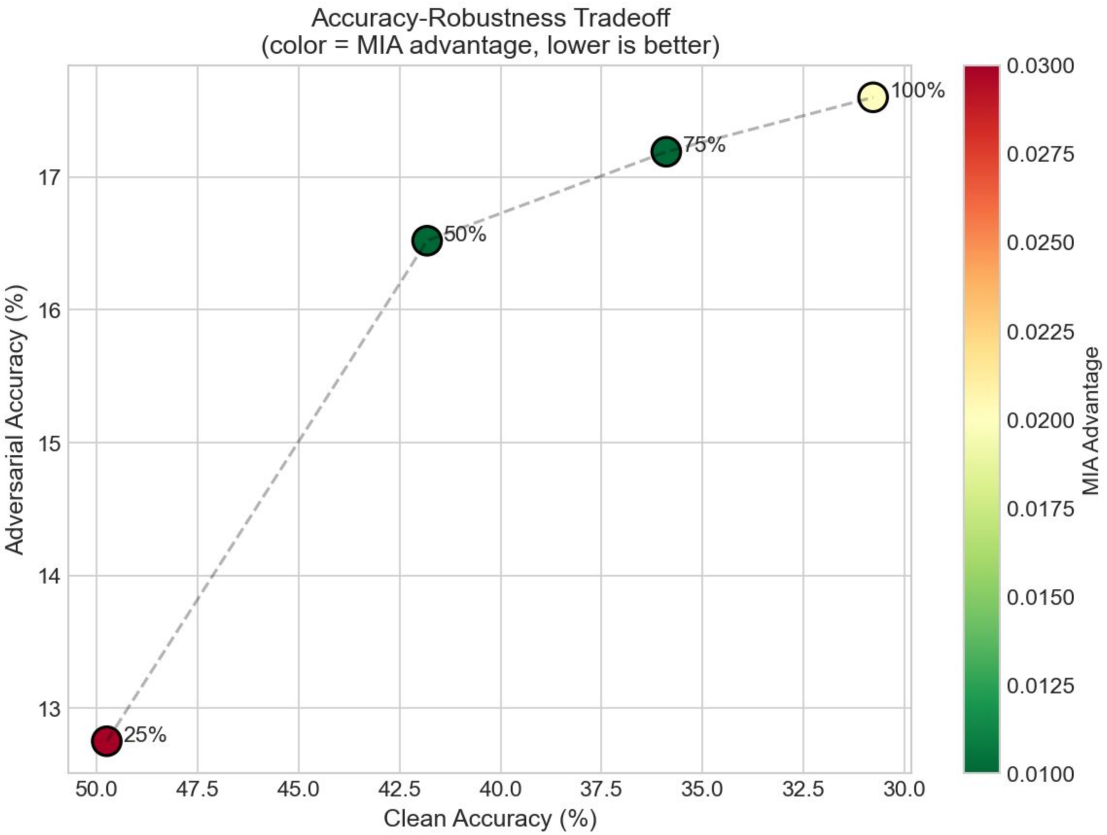
Attack	Part 1 Advantage	Part 2 Advantage	Reduction
Confidence Threshold	0.305	0.018	94.1%
Logits Threshold	0.309	0.027	91.3%
Loss Threshold	0.312	0.006	98.1%
Gap Attack	0.145	-0.014	109.7%
Augmentation	0.174	-0.023	113.2%

Table 7: MIA advantage comparison. Part 2's adversarially trained model reduces all attack advantages to near-zero.

Mix adversarial training



Accuracy-Robustness Tradeoff



Conclusion

- **Without retraining - stronger privacy, constrained robustness**

- **Adversarial training provides strong robustness and privacy**

- **Accuracy-Robustness Trade-offs [7]**

References

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